No.	Status	Date
LNDE-RPT-20140902-024-DS-4-RA-104	Issued	2014-09-19

UT Thickness Data Sheet - 4-RA-104

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for line 4-RA-104 located within the ATU-100. The scope of work involves a pre-change rich-amine survey to establish current wall thickness and evaluate asset integrity prior to the planned Modification Order.

Marcus Chen, acting as the lead UT technician and SNT-TC-1A UT Level II for Larkhall NDE Services Inc., is responsible for the execution of the measurements and the recording of the data presented herein.

Technical Basis and Procedures

All thickness measurements were performed under the requirements of QSCP-NDT-001 Rev 2. This procedure incorporates specific requirements for the CUI program, including the use of inspection ports and the management of insulation removal and reinstatement.

The measurement techniques and calibration standards are cited in accordance with ASTM E797. The data collected in this sheet supports the findings detailed in the primary report LNDE-RPT-20140902-024-R0.

Component Specifications

The subject line, 4-RA-104, is the rich side of the L/R exchanger. The material of construction is A106 Gr.B with a nominal outer diameter of 114.3 mm and a nominal wall thickness of 8.56 mm.

The design parameters for this circuit are established at a maximum allowable pressure of 862 kPa(g) and a maximum allowable temperature of 127 °C. The minimum allowable thickness (t-min) for this line is 3.81 mm.

Inspection Context

This survey is part of the 3-year interval set by the DMR for rich-amine service. The previous thickness readings for this circuit were recorded in 2011-02.

The current data is utilized to determine the corrosion rate and remaining life by comparing the 2014-09-19 readings against the 2011-02 baseline, ensuring the line remains above the retirement thickness of 3.81 mm.

Thickness readings, measured 2014-09-19

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2011-02-18	8.32	8.07	0.024	0.070	>50	2017-09-19	
4-RA-101-02	8.56	3.81	2011-02-18	8.20	8.02	0.028	0.050	>50	2017-09-19	
4-RA-101-03	8.56	3.81	2011-02-18	7.98	7.88	0.039	0.028	>50	2017-09-19	
4-RA-101-04	8.56	3.81	2011-02-18	8.21	8.18	0.024	0.008	>50	2017-09-19	
4-RA-101-05	8.56	3.81	2011-02-18	8.16	7.98	0.033	0.050	>50	2017-09-19	
4-RA-101-06	8.56	3.81	—	—	7.91	—	—	—	2017-09-19	
4-RA-101-07	8.56	3.81	2011-02-18	8.30	8.22	0.021	0.022	>50	2017-09-19	
4-RA-101-08	8.56	3.81	2011-02-18	8.02	7.71	0.052	0.086	45.1	2017-09-19	
4-RA-101-09	8.56	3.81	2011-02-18	8.32	8.27	0.016	0.014	>50	2017-09-19	
4-RA-101-10	8.56	3.81	2011-02-18	8.28	8.21	0.025	0.020	>50	2017-09-19	
4-RA-101-11	8.56	3.81	2011-02-18	8.04	7.88	0.043	0.045	>50	2017-09-19	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2011-02-18	8.33	8.25	0.016	0.022	>50	2017-09-19	
4-RA-102-02	8.56	3.81	2011-02-18	7.99	7.89	0.037	0.028	>50	2017-09-19	
4-RA-102-03	8.56	3.81	2011-02-18	8.04	7.95	0.033	0.025	>50	2017-09-19	
4-RA-102-04	8.56	3.81	2011-02-18	7.96	7.93	0.033	0.008	>50	2017-09-19	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-05	8.56	3.81	2011-02-18	8.07	7.84	0.038	0.064	>50	2017-09-19	
4-RA-102-06	8.56	3.81	2011-02-18	8.01	7.95	0.030	0.017	>50	2017-09-19	
4-RA-102-07	8.56	3.81	2011-02-18	8.10	7.98	0.032	0.033	>50	2017-09-19	
4-RA-102-08	8.56	3.81	2011-02-18	7.88	7.58	0.049	0.084	45.0	2017-09-19	
4-RA-102-09	8.56	3.81	2011-02-18	7.98	7.84	0.038	0.039	>50	2017-09-19	
4-RA-102-11	8.56	3.81	2011-02-18	8.19	8.12	0.020	0.020	>50	2017-09-19	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2011-02-18	10.47	10.62	0.021	-0.042	>50	2017-09-19	
6-RA-103-02	10.97	4.57	2011-02-18	10.23	10.06	0.056	0.047	>50	2017-09-19	
6-RA-103-03	10.97	4.57	2011-02-18	10.42	10.39	0.032	0.008	>50	2017-09-19	
6-RA-103-04	10.97	4.57	2011-02-18	9.97	9.79	0.064	0.050	>50	2017-09-19	
6-RA-103-05	10.97	4.57	2011-02-18	10.10	9.81	0.062	0.081	>50	2017-09-19	
6-RA-103-06	10.97	4.57	2011-02-18	10.56	10.43	0.031	0.036	>50	2017-09-19	
6-RA-103-07	10.97	4.57	2011-02-18	10.55	10.37	0.033	0.050	>50	2017-09-19	
6-RA-103-08	10.97	4.57	2011-02-18	10.44	10.49	0.026	-0.014	>50	2017-09-19	
6-RA-103-09	10.97	4.57	2011-02-18	10.56	10.39	0.033	0.047	>50	2017-09-19	
6-RA-103-10	10.97	4.57	2011-02-18	10.05	9.93	0.059	0.033	>50	2017-09-19	
6-RA-103-11	10.97	4.57	—	—	10.10	—	—	—	2017-09-19	
6-RA-103-13	10.97	4.57	2011-02-18	10.45	10.38	0.020	0.020	>50	2017-09-19	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2011-02-18	7.95	7.84	0.036	0.031	>50	2017-09-19	
4-RA-104-02	8.56	3.81	2011-02-18	8.14	8.09	0.025	0.014	>50	2017-09-19	
4-RA-104-03	8.56	3.81	2011-02-18	8.56	8.46	0.032	0.028	>50	2017-09-19	
4-RA-104-04	8.56	3.81	2011-02-18	8.60	8.49	0.007	0.031	>50	2017-09-19	
4-RA-104-05	8.56	3.81	2011-02-18	8.57	8.37	0.015	0.056	>50	2017-09-19	
4-RA-104-06	8.56	3.81	2011-02-18	7.98	7.91	0.037	0.020	>50	2017-09-19	
4-RA-104-07	8.56	3.81	2011-02-18	7.78	7.61	0.053	0.047	>50	2017-09-19	
4-RA-104-08	8.56	3.81	2011-02-18	8.18	8.00	0.050	0.050	>50	2017-09-19	

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT Level II	2014-09-19	_____